

GE261004 DESIGN THINKING

Unit III – Ideation and Creative Thinking · Presentation 5 of 6

Lateral Thinking and the Six Thinking Hats

Vertical and lateral thinking · four lateral tools · provocation and movement · de Bono's six hats · why parallel thinking works

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In this presentation

- Vertical and lateral thinking
- Digging a different hole
- Four lateral tools
- Provocation and movement
- The six thinking hats
- What each hat does
- Running a Six Hats session
- Why parallel thinking works

1

Lateral thinking

Changing where you are looking, not how hard

Vertical and lateral thinking

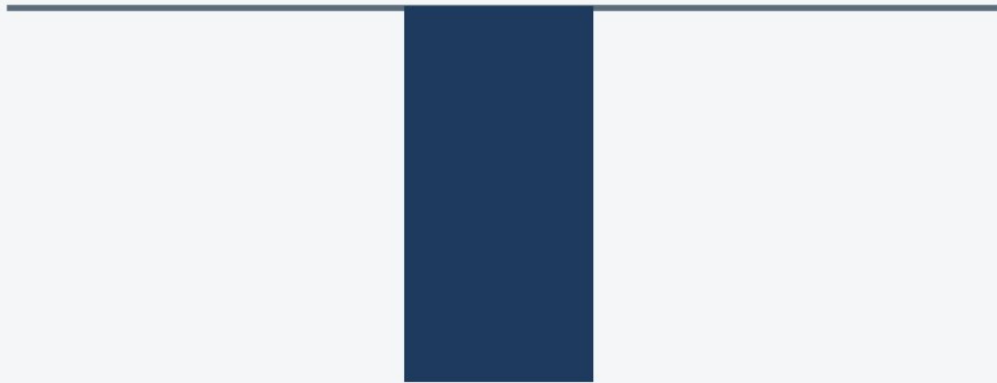
- Edward de Bono coined the term in 1967 for thinking that moves sideways, not downwards
- **Vertical thinking** is linear and methodical: it develops one pattern further
- **Lateral thinking** deliberately breaks the pattern to make a different one possible
- Both are necessary – lateral finds the direction, vertical does the work

KEY POINT Logic will dig a hole deeper. It will not tell you the hole is in the wrong place.

Digging a different hole

VERTICAL THINKING

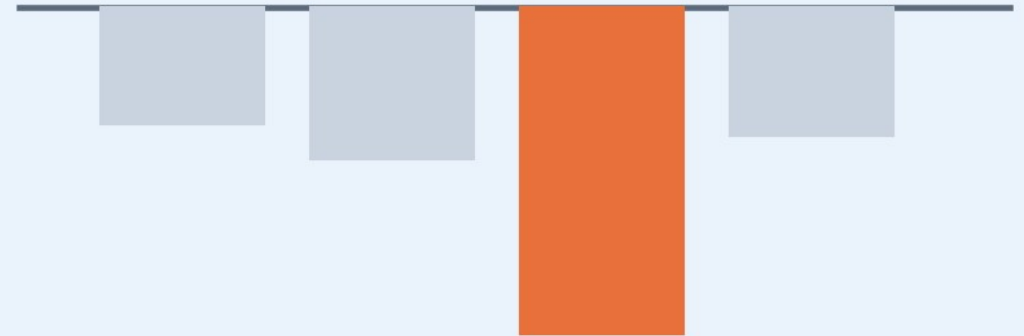
dig the same hole deeper



one idea, pushed hard

LATERAL THINKING

dig somewhere else



many starting points, one pays off

You cannot reach the second hole by improving the first one.

Four lateral tools

Alternatives

Deliberately generate a fixed number of other ways, even when the first one already works.

Challenge

Ask why it is done this way at all. Not because it is wrong – because nobody has asked.

Provocation

State something deliberately unreasonable, then look for what it makes possible.

Random entry

Take an unrelated word and force a connection. The unrelatedness is the point.

Provocation and movement

- A provocation is a statement you do not believe: “cars have square wheels”
- You do not judge it – you move from it, asking what it would make true
- Square wheels need a suspension that anticipates the road, which is a real idea
- De Bono's signal word for this is PO: a marker meaning “this is a provocation, do not argue”

KEY POINT Judgement asks “is this true?”. Movement asks “where does this take me?”
Only the second one generates.

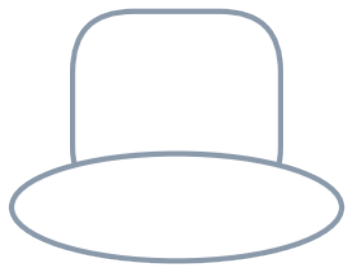
2

The Six Thinking Hats

Everyone looking the same way at the same time

The six thinking hats

Everyone wears the same hat at the same time



WHITE

facts



RED

feelings



BLACK

caution



YELLOW

benefits



GREEN

new ideas



BLUE

process

parallel thinking: the team looks in one direction, then turns together

De Bono published the method in 1985; the hats are roles, not personalities.

What each hat does

Hat	The mode	What the room does
White	Facts	States what is known and what is missing
Red	Feelings	Says gut reactions, with no justification
Black	Caution	Names risks and what could go wrong
Yellow	Benefits	Names value and why it could work
Green	New ideas	Generates alternatives and proposals
Blue	Process	Runs the session and decides the order

Running a Six Hats session

- Blue hat first: agree the question and the order of the other hats
- Everyone wears the same hat at the same time, for a fixed few minutes
- A typical order: Blue, White, Green, Yellow, Black, Red, Blue
- Black last among the judging hats – caution early kills the green hat

ACTIVITY Eighteen minutes: three minutes a hat on your team's chosen idea. The Blue hat keeps time and nobody argues out of turn.

Why parallel thinking works

- In an ordinary argument, two people look in opposite directions and defend positions
- With hats, everyone looks the same way at once, then the whole room turns together
- Nobody has to be the pessimist: the Black hat is a role, not a personality
- It gives the cautious person permission to be optimistic, and the reverse

KEY POINT The hats separate the thinking from the thinker. That is what stops a review turning into a fight.

In class: hats on a real idea

- Each team puts one idea from the SCAMPER session on the board
- Blue sets the order; the whole team wears each hat together
- One person records under six headings – nothing is discarded
- End on Blue: what did the hats change about the idea?

ACTIVITY Twenty minutes. The recorder reads back only what changed, in one minute.

THINK ABOUT IT

Which hat does your team wear by default – and which one never gets worn?

- Who in your team is always the Black hat, and is that fair to them?
- When did caution last kill an idea before it was understood?
- What would happen if you ran Red first for once?
- Which hat would most improve your next review?

Summary

- De Bono coined lateral thinking in 1967: change direction, not effort
- Four tools – alternatives, challenge, provocation, random entry
- A provocation is moved from, never argued with
- Six Thinking Hats, 1985: facts, feelings, caution, benefits, ideas, process
- Everyone wears the same hat at the same time
- The hats separate the thinking from the thinker

Next: Concept selection and evaluation

- Converging as a sequence of screens
- Clustering, dot voting and quick screens
- The three-lens test applied to concepts
- The Pugh matrix and how to read it